

Experiment no:6

Determination and plotting of Capacitance of parallel plate capacitor in three different Dielectric medium with specific area, dielectric thickness and various dielectric constant.

Aim:

To write program for capacitance of parallel plate capacitor in three different dielectric media using SCILAB.

Software required:

- SCILAB version 6.0.1

Program:

```
EX 6.sci (C:\Users\user\Documents\EX 6.sci) - SciNotes
File Edit Format Options Window Execute ?
EX 6.sci (C:\Users\user\Documents\EX 6.sci) - SciNotes
EX 6.scd X

1  clc;
2  clear;
3
4  // Parallel-Plate Capacitor
5  a1 = input("Enter the a1 value is:");
6  d = input("Enter the d value is:");
7
8  e = 8.854 * 10^-12;
9  c = e * a1 / d;
10
11 disp("Capacitance of Parallel-Plate-Capacitor:");
12 disp(c);
13
14 // Plot
15 d_plot = linspace(0.1, 10, 50); // Start from 0.1 to avoid division by zero
16 y = c * exp(-0.1 * d_plot);
17
18 plot(d_plot, y);
19 xlabel("Distance (d)");
20 ylabel("Capacitance");
21 title("Parallel-Plate-Capacitor");
22
23 // Multi-dielectric Capacitor
24 a2 = input("Enter the a2 value is:");
25 d1 = input("Enter the d1 value is:");
26 d2 = input("Enter the d2 value is:");
27 d3 = input("Enter the d3 value is:");
28
29 er1 = input("Enter the er1 value is:");
30 er2 = input("Enter the er2 value is:");
31 er3 = input("Enter the er3 value is:");
32
33 c1 = e * a2 / ((d1 / er1) + (d2 / er2) + (d3 / er3));
34
35 disp("Capacitance of Multi-dielectric-Capacitor:");
36 disp(c1);
37
```

Output:

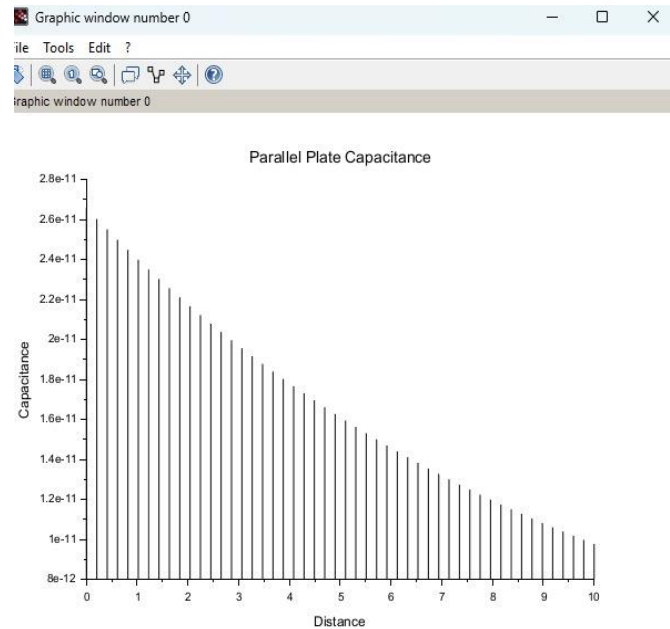
```

Enter the a1 value is: 0.0003
Enter the d value is: 0.001

"Capacitance of Parallel Plate Capacitor:"
2.656D-12
Enter the a2 value is: 0.0006
Enter the d1 value is: 0.002
Enter the d2 value is: 0.003
Enter the d3 value is: 0.004
Enter the er1 value is: 4
Enter the er2 value is: 6
Enter the er3 value is: 8

"Capacitance of Multi-dielectric Capacitor:"
3.542D-12

```



Theoretical calculation :

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Output :

Enter the a1 value is : 3 cm²

Enter the d value is : 1 mm

Enter the a2 value is : 6 cm²

Enter the d1 value is : 2 mm

Enter the d2 value is : 3 mm

Enter the d3 value is : 4 mm

Enter the er1 value is : 4

Enter the er2 value is : 6

Enter the er3 value is : 8

Theoretical output :

"Capacitance of parallel plate capacitor :

2.656D-12

"Capacitance of Multi-dielectric capacitor :

3.542D-12

Theoretical calculation :

$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$

$C = \frac{\epsilon A}{d}$

$A = 3 \text{ cm}^2 = 3 \times 10^{-4} \text{ m}^2$

$d = 1 \text{ mm} = 1 \times 10^{-3} \text{ m}$

$C = \frac{8.854 \times 10^{-12} \times 3 \times 10^{-4}}{1 \times 10^{-3}}$

$C = 2.6562 \times 10^{-12} \text{ F}$

$C = 2.6562 \times 10^{-12} \text{ F}$

$C = 2.6562 \text{ pF}$

Capacitance of parallel plate capacitor

$C = 2.6562 \text{ pF}$

Capacitance of each dielectric

Dielectric 1 $\epsilon_r = 4$

$d_1 = 2 \text{ mm}$

$C_1 = \frac{8.854 \times 10^{-12} \times 4 \times 3 \times 10^{-4}}{2 \times 10^{-3}}$

$C_1 = 5.3124 \times 10^{-12} \text{ F}$

$C_1 = 5.3124 \text{ pF}$

Dielectric 2 $\epsilon_r = 6$

$d_2 = 3 \text{ mm}$

$C_2 = \frac{8.854 \times 10^{-12} \times 6 \times 3 \times 10^{-4}}{3 \times 10^{-3}}$

$C_2 = 5.3124 \times 10^{-12} \text{ F}$

$C_2 = 5.3124 \text{ pF}$

Dielectric 3 $\epsilon_r = 8$

$d_3 = 4 \text{ mm}$

$C_3 = \frac{8.854 \times 10^{-12} \times 8 \times 3 \times 10^{-4}}{4 \times 10^{-3}}$

$C_3 = 5.3124 \times 10^{-12} \text{ F} = 5.3124 \text{ pF}$

Case 1. AIR:

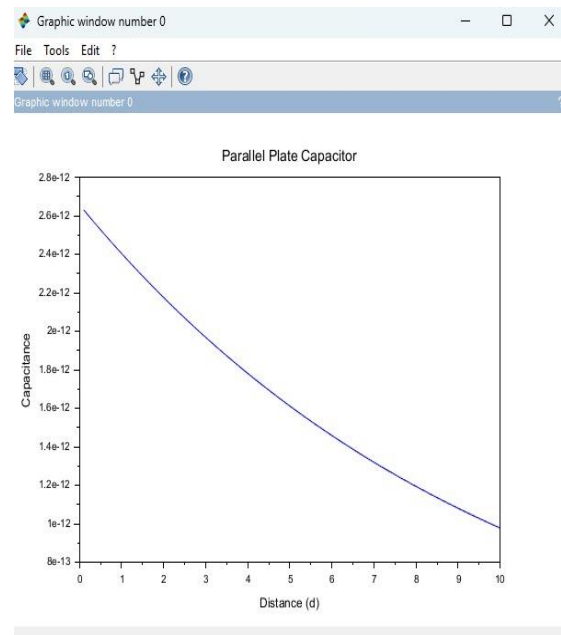
```
4 //Plot
5 d_plot = linspace(0.1, 10, 50); %//Start from 0.1 to avoid division by zero
6 y = c * exp(-0.1 * d_plot);
7
8 plot(d_plot, y);
9 xlabel("Distance (d)");
10 ylabel("Capacitance");
11 title("Parallel-Plate-Capacitor");
12
13 //Multi-dielectric-Capacitor
14 a2 = input("Enter the a2 value is:");
15 d1 = input("Enter the d1 value is:");
16 d2 = input("Enter the d2 value is:");
17 d3 = input("Enter the d3 value is:");
18
19 er1 = input("Enter the er1 value is:");
20 er2 = input("Enter the er2 value is:");
21 er3 = input("Enter the er3 value is:");
22
23 c1 = e * a2 / ((d1 / er1) + (d2 / er2) + (d3 / er3));
24
25 disp("Capacitance of Multi-dielectric Capacitor:");
26 disp(c1);
```

Output:

```
Enter the a1 value is: 0.0003
Enter the d value is: 0.001

"Capacitance of Parallel Plate Capacitor:"
2.656D-12
Enter the a2 value is: 0.0006
Enter the d1 value is: 0.002
Enter the d2 value is: 0.003
Enter the d3 value is: 0.004
Enter the er1 value is: 4
Enter the er2 value is: 6
Enter the er3 value is: 8

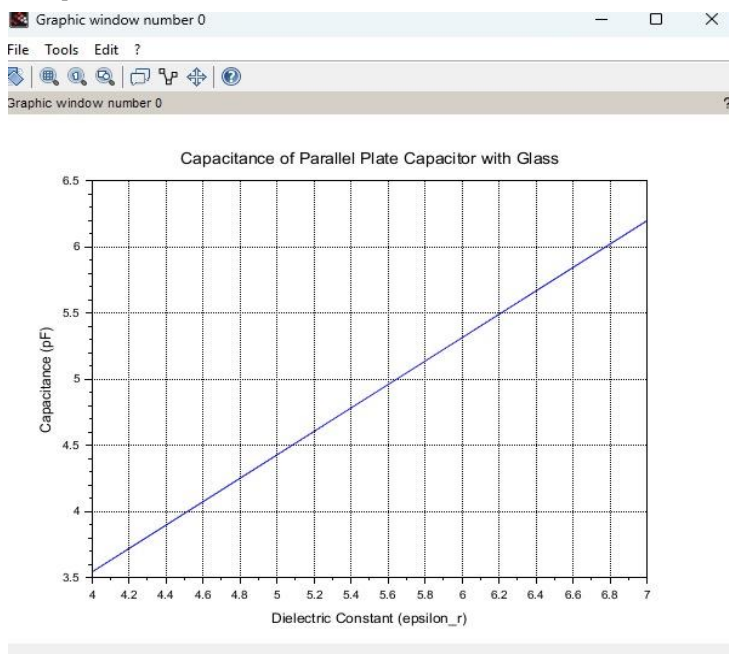
"Capacitance of Multi-dielectric Capacitor:"
3.542D-12
```



Case 2. Glass:

```
EX 6.sci (C:\Users\user\Documents\EX 6.sci) - SciNotes
File Edit Format Options Window Execute ?
[Icons]
EX 6.sci (C:\Users\user\Documents\EX 6.sci) - SciNotes
EX 6.sci X
1 clc;
2 clear;
3
4 //Constants
5 epsilon_0 = 8.854e-12;
6 area = 1e-4; //m^2
7 thickness = 1e-3; //m
8
9 //Dielectric constant of Teflon
10 epsilon_r_teflon = linspace(2.0, 2.5, 100);
11
12 //Capacitance calculation
13 capacitance_teflon = (epsilon_r_teflon .* epsilon_0 .* area) ./ thickness;
14
15 //Plot graph
16 plot(epsilon_r_teflon, capacitance_teflon * 1e12);
17
18 xlabel("Dielectric Constant (epsilon_r)");
19 ylabel("Capacitance (pF)");
20 title("Capacitance of Parallel Plate Capacitor with Teflon");
21 xgrid();
```

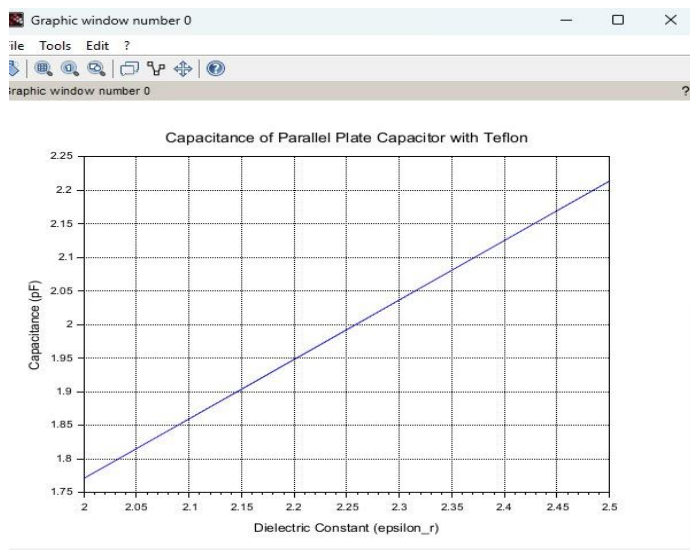
Output:



Case 3. Teflon:

```
EX 6.sci (C:\Users\user\Documents\EX 6.sci) - SciNotes
File Edit Format Options Window Execute ?
EX 6.sci (C:\Users\user\Documents\EX 6.sci) - SciNotes
EX 6.sci X
1 clc;
2 clear;
3
4 // Constants
5 epsilon_0 = 8.854e-12;
6 area = 1e-4; //m^2
7 thickness = 1e-3; //m
8
9 // Dielectric constant of glass
10 epsilon_r_glass = linspace(4.0, 7.0, 100);
11
12 // Capacitance calculation
13 capacitance_glass = (epsilon_r_glass .* epsilon_0 .* area) ./ thickness;
14
15 // Plot
16 plot(epsilon_r_glass, capacitance_glass * 1e12);
17
18 xlabel("Dielectric Constant (epsilon_r)");
19 ylabel("Capacitance (pF)");
20 title("Capacitance of Parallel Plate Capacitor with Glass");
21 xgrid();
22
```

Output:



Result :

Thus the program was verified for capacitance of parallel plate capacitor in three different dielectric media using SCILAB was successfully.